Math 241, F04
Exam 2

Name:
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	15	
2	15	
3	15	
4	20	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (15 points) Given a function

$$f(x,y) = e^{x^2-y^2},$$

a) Find its two first partial derivatives $f_x(x,y)$ and $f_y(x,y)$.

Solution:

$$\begin{aligned} f_x &= 2xe^{x^2-y^2} \\ f_y &= -2ye^{x^2-y^2} \end{aligned}$$

b) Find its four second partial derivatives $f_{xx}(x,y)$, $f_{yy}(x,y)$, $f_{xy}(x,y)$, $f_{yx}(x,y)$.

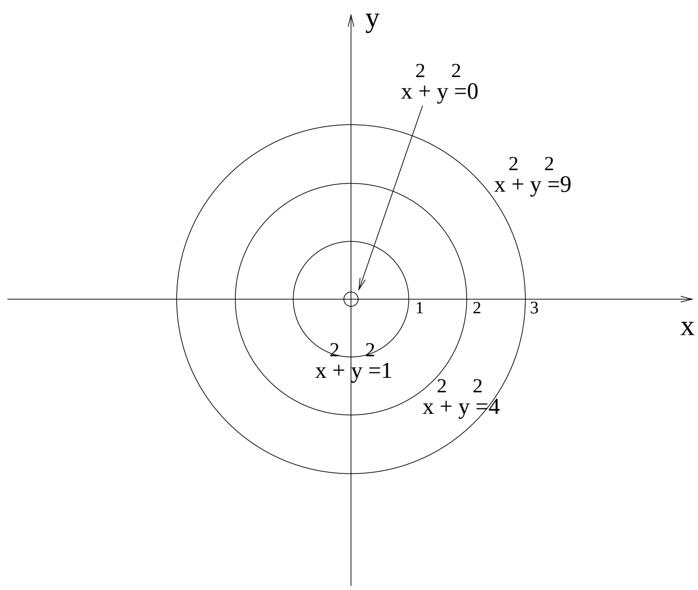
Solution:

$$\begin{aligned} f_{xx} &= 2e^{x^2-y^2} + 2xe^{x^2-y^2}(2x) = (4x^2+2)e^{x^2-y^2} \\ f_{yy} &= -2e^{x^2-y^2} - 2ye^{x^2-y^2}(-2y) = (4y^2-2)e^{x^2-y^2} \\ f_{xy} &= 2xe^{x^2-y^2}(-2y) = -4xye^{x^2-y^2} \\ f_{yx} &= -2ye^{x^2-y^2}(2x) = -4xye^{x^2-y^2} \end{aligned}$$

Problem 2. (15 points)

a) Let $z = x^2 + y^2$, sketch the level curves of $z = k$ for $k = 0, 1, 4, 9$ in the plane. (*Remember to label the curves and the axes.*)

Solution:



b) Find the indicated limit or state that it doesn't exit.

$$(1) \qquad \lim_{(x,y)\to(\mathbf{1},\mathbf{1})} \frac{x^2 - y^2}{x^4 - y^4}$$

Solution:

$$= \lim_{(x,y)\to(\mathbf{1},\mathbf{1})} \frac{x^2 - y^2}{(x^2 - y^2)(x^2 + y^2)} = \lim_{(x,y)\to(\mathbf{1},\mathbf{1})} \frac{1}{(x^2 + y^2)} = \frac{1}{2}$$

$$(2) \qquad \lim_{(x,y)\to(\mathbf{0},\mathbf{0})} \frac{x^2 - y^2}{x^4 - y^4}$$

Solution:

$$= \lim_{(x,y)\to(\mathbf{1},\mathbf{1})} \frac{x^2 - y^2}{(x^2 - y^2)(x^2 + y^2)} = \lim_{(x,y)\to(\mathbf{1},\mathbf{1})} \frac{1}{(x^2 + y^2)} = \frac{1}{0}$$

So it doesn't exist.

Problem 3. (15 points)

a) Let $f(x,y) = \dfrac{x^2}{y}$, find the gradient vector ∇f at $(-2,3)$.

Solution:

$$f_x = \frac{2x}{y}$$
$$f_y = -\frac{x^2}{y^2}$$

So

$$\nabla f(-2,3) =< -\frac{3}{4}, -\frac{4}{9} >$$

b) Given $f(x,y) = x^2 - 4xy + 2y^2$, $\mathbf{p} = (-1,2)$ and $\mathbf{a} = \mathbf{i} - \sqrt{3}\mathbf{j}$, find the directional derivative of f at the point $\mathbf{p}$ in the direction of $\mathbf{a}$.

Solution: The unit vector $\mathbf{u}$ in the directiopn of $\mathbf{a}$ is:

$$\mathbf{u} = \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{\mathbf{i} - \sqrt{3}\mathbf{j}}{2} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{\sqrt{3}}{2}\mathbf{j}$$

We also know

$$f_x(x,y) = 2x - 4y, \quad f_y(x,y) = -4x + 4y,$$

thus,

$$f_x(-1,2) = -10, \quad f_y(-1,2) = 12.$$

Consequently, we have

$$D_{\mathbf{u}}(3,-2) = -10 \cdot \frac{1}{\sqrt{2}} + 12 \cdot (-\frac{\sqrt{3}}{2}) = -5 - 6\sqrt{3}$$

Problem 4. (20 points)

a) Given a surface determined by $xe^{-2y} - 2z = 0$, find the equation of the tangent plane to this surface at the point $(2, 0, 1)$.

Solution: Let $F(x, y, z) = xe^{-2y} - 2z$, then

$$F_x(x, y, z) = e^{-2y}, \quad F_y(x, y, z) = -xe^{-2y}, \quad F_z(x, y, z) = -2,$$

then

$$F_x(2, 0, 1) = 1, \quad F_y(2, 0, 1) = -4, \quad F_z(2, 0, 1) = -2.$$

So the equation of the tangent plane at $(2, 1, 1)$ is

$$\begin{aligned} (x-2) - 4(y-0) - 2(z-1) &= 0 \\ x - 4y - 2z &= 0 \end{aligned}$$

b) Let $w = x^2 \sin y$, $x = st$, $y = s - t$, find $\dfrac{\partial w}{\partial t}$ using chain rule and express the final answer in term of s and t.

Solution:

$$\begin{aligned} \frac{\partial w}{\partial x} &= 2x \sin y, \quad \frac{\partial x}{\partial t} = s \\ \frac{\partial w}{\partial y} &= x^2 \cos y, \quad \frac{\partial y}{\partial t} = -1 \end{aligned}$$

So we have

$$\begin{aligned} \frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} \\ &= 2x \sin y \cdot s + x^2 \cos y \cdot (-1) \\ &= 2s^2 t \sin(s-t) - (st)^2 \cos(s-t) \end{aligned}$$

Problem 5. (20 points) Given a function

$$f(x,y) = x^2y - 6y^2 - 3x^2,$$

a) Find all critical points of f.

Solution: Let $\nabla f = 0$, then we have

$$\begin{aligned} f_x &= 2xy - 6x = 0 \qquad (1) \\ f_y &= x^2 - 12y = 0 \qquad (2) \end{aligned}$$

From (2), we get $y = x^2/12$ and substitute it in (1), then obtain

$$\begin{aligned} 2x(\frac{x^2}{12}) - 6x &= 0 \\ x^3 - 36x &= 0 \\ x(x^2 - 36) &= 0 \end{aligned}$$

Thus, $x = 0$, $x = 6$ or $x = -6$. Clearly $y = 0$ when $x = 0$ and $y = 3$ when $x = 6$ or -6. Therefore, the critical points are $(0,0)$, $(6,3)$ and $(-6,3)$.

b) Determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point using the **second partials test**.

Solution: $f_{xx} = 2y - 6$, $f_{yy} = -12$, and $f_{xy} = 2x$, so

$$\begin{aligned} D = D(x,y) &= f_{xx}f_{yy} - f_{xy}^2 \\ &= (2y-6)(-12) - (2x)^2 \\ &= -4x^2 - 24y + 72 \end{aligned}$$

For the critical point $(0,0)$, $D(0,0) = 72 > 0$ and $f_{xx}(0,0) = -6 < 0$, so $f(0,0) = 0$ gives a local maximum.

For the critical point $(6,3)$, $D(6,3) = -4(6)^2 - 24 \cdot 3 + 72 = -144 < 0$, so $(6,3)$ is a saddle point.

For the critical point $(-6,3)$, $D(-6,3) = -4(-6)^2 - 24 \cdot 3 + 72 = -144 < 0$, so $(-6,3)$ is a saddle point.

Problem 6. (15 points) Find the minimum of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $x + 3y - 2z - 12 = 0$ using **Lagrange's method**.

Solution: First we have

$$\nabla f(x, y, z) =< 2x, 2y, 2z > .$$

Let $g = x + 3y - 2z - 12 = 0$, then

$$\nabla g(x, y) =< 1, 3, -2 > .$$

The Lagrange equztion becomes

$$\begin{aligned} 2x &= \lambda && (1) \\ 2y &= 3\lambda && (2) \\ 2z &= -2\lambda && (3) \\ x + 3y - 2z - 12 &= 0 && (4) \end{aligned}$$

Let (1), (2) and (3), we could find

$$x = \lambda/2, \quad y = 3\lambda/2, \quad z = -\lambda$$

Substitute them in (4), we get

$$\begin{aligned} \lambda/2 + 3 \cdot (3\lambda/2) - 2(-\lambda) &= 12 \\ 7\lambda &= 12 \\ \lambda &= 7/12 \end{aligned}$$

Thus, $x = 6/7$, $y = 18/7$ and $z = -12/7$. Therefore the minimum value of f subject to the constraint is

$$f(6/7, 18/7, -12/7) = (6/7)^2 + (18/7)^2 + (-12/7)^2 = 504/49 = 72/7.$$